ManimAgent: Self-Evolving Multimodal Agents for Visual Education

Wenjia Jiang · Zongyuan Cai · Yuanhang Shao · Chenru Wang · Boyan Han · Zhixue Song · Keyu Chen · Shengwei An · Xu Yang · Zhou Yang
arXiv 2606.30296v2 · Independent Reel interpretation

HarnessRSI 67 ★★

evidence: retained

Paper

Code

1 What changes and what stays

An external dual-channel memory bank grows across tasks. No weight updates or human seeds.

1 · Retrieve before coding
2 success examples → soft guidance
3 failure lessons → Known Pitfalls

↓

2 · Generate and render
GPT-5.5 Coder + sandbox checks
Crash repair: at most 2 retries

↓

3 · Review, revise, select
GPT-5.5 judges 4 keyframes
2 visual revisions; auto-pass at 90

↓

Keep highest-score render
M+ · best visual score ≥ 85
Success rationale + chosen code
M− · crash → render success,
or adjacent visual gain ≥ 5
Distilled cause + fix + examples

↳ **Persistent bank → next task**

§3; Algorithm 1. “Hard” pitfalls are prompt instructions. No eviction or paraphrase dedup.

2 Strong RAG closes most of the gap

Human Pass@1 (%) · individual usable votes

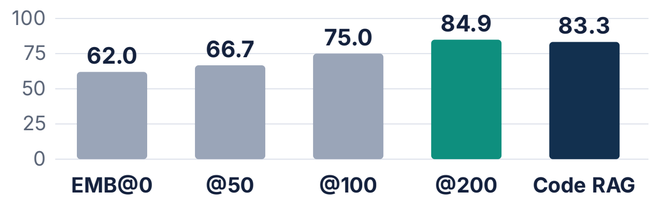


Table 1	Pass %	Rounds ↓	Quality
Full EMB	**84.9**	**6.5**	**3.88**
Code RAG	83.3	11.4	3.65

GPT-5.5; 39 build → 33 paper-disjoint probe tasks, read-only. Human subset: 25 tasks, 54 blind raters. Point estimates; no run variance or CIs.

4 Where the gain is small or uncertain

- Only **EMB@200** exceeds code RAG on human metrics. @50 / @100 remain below it.
- Top-1 quality **3.84** is close to full **3.88**. No per-task/domain gain breakdown.
- **Human agreement:** binary κ = −0.01 (153 videos), 59.4% unanimous. Likert ICC also low (23 videos). Denominators unresolved.
- **Reward risk:** GPT-5.5 revises and gates writes; VLM–human r = −0.17 (37 videos). Blind ratings mitigate favoritism.

3 Controls, budgets and missing tests

Table 1 variant	Pass %	Rounds ↓	Quality
Full EMB@200	**84.9**	**6.5**	**3.88**
No success M+	70.0	12.0	3.40
No failure M−	80.9	11.6	3.52
No memory*	61.8	12.2	3.20
Rationale only	75.0	10.5	3.50
Ungated failure	81.8	8.4	3.64
Top-1 / channel	81.0	9.9	3.84
Weak reviewer	80.0	8.6	3.42
Doc-only RAG	82.1	11.5	3.57

*Separate ablation, not EMB@0. Weak reviewer: Step-3.7-Flash. All rows lack run uncertainty.

- **Matched retrieval:** RAG shares encoder, index, depth, truncation and reflection limits. Random EMB shuffles content. No matched total build-cost control.
- **Fewer rounds ≠ faster:** full EMB 31 min vs RAG 27; estimated LLM/VLM tokens 36K/20K vs 38K/23K. No explicit token tracking or total bank-build bill.
- **Defaults:** fixed across conditions; selection split and baseline tuning budgets not stated.
- **Safeguards:** paper/field isolation validator, read-only probe, cutoff stratification. Cross-test is qualitative only; no quantitative cross-model/renderer transfer.

Takeaway

Retained success examples and repair lessons help later animation tasks. Against strong static RAG, human gains are modest and unquantified statistically; fewer repairs still cost more rendering time. This is memory reuse, not demonstrated recursive improver evolution.

Our read · HarnessRSI

67 · ★★ · evidence: retained

Reports: disjoint frozen probe, channel/gate ablations, human ratings, estimated task costs.
Not reported: seed variance / CIs, total build cost, quantitative transfer.